

Data Science Capstone Project EDA and Feature Engineering

Course: MSDS640 Data Science Capstone · Dataset: UCI Adult Census Income (1994 U.S. Census) · Erik Herb

The target is income, either over 50K or 50K and under a year, and I explore the data with my three hypotheses in mind rather than plotting for the sake of plotting. Hypothesis 1 is that education is a stronger driver of the over 50K prediction than hours worked. Hypothesis 2 is that XGBoost is the fairest of the three models toward true high earners across nativity, with a smaller US-born versus foreign-born recall gap than logistic regression and the MLP. Hypothesis 3 is that XGBoost classifies better than both the logistic baseline and the MLP on PR-AUC and F1. As in the first milestone, the data is loaded into SQLite and pulled back out before any analysis.

Two things are new in this version. My professor suggested using a decision tree to estimate feature importance and to help explain how a model makes its decisions, so section 7.1 does that for Hypothesis 1 and section 8 is the explainability note for the final report's model comparison section. My professor also suggested adding a t-test, so section 7.2 puts statistical tests behind the EDA patterns, chosen to match what each feature actually is.

1. Handling missing values

Only three columns are missing anything: workclass at 5.64 percent, occupation at 5.66 percent, and native country at 1.79 percent. All three are categorical and well under the level where dropping a column would make sense. I fill each with an explicit Unknown category rather than dropping the rows, because dropping every row with a missing value would throw away roughly 2,400 people, and the missingness is not random, so dropping them could bias the income analysis. An Unknown bucket keeps the rows and lets the model treat missingness as its own signal.

I also drop the 24 exact duplicate rows while the sampling weight is still in the table, so genuinely different people are not merged by mistake. I then drop that sampling weight, which is not a fact about the person, and the text education column, which repeats the numeric one. After cleaning there are no missing values left.

2. Addressing data imbalance

The target is about 75.9 percent under 50K and 24.1 percent over 50K, roughly a 3.2 to 1 split. That drives both my choice of metrics and how I handle the imbalance.

On metrics, plain accuracy is misleading here, because a model that predicts nobody earns over 50K is right about 76 percent of the time while catching zero high earners. Precision is the share of predicted high earners who really are high earners, and recall is the share of real high earners the model actually catches. F1 combines those two, so it is more honest than accuracy when the classes are uneven. ROC-AUC can still look flattering when the positive class is small, because a large easy negative class props up the curve. PR-AUC, the area under the precision-recall curve, focuses on the positive class directly and does not get credit for the easy negatives, so it is the most reliable single number for an imbalanced problem like this. That is why my plan leads with PR-AUC and recall and treats accuracy as a secondary number.

On handling, I plan to train with class weights and with SMOTE oversampling on the training split only, and compare both against the raw imbalance. I prefer oversampling the minority class over undersampling the majority, because undersampling would throw away most of the under 50K rows and lose real information, while oversampling keeps every real row. I keep the oversampling on the training split only so the validation and test sets stay a real reflection of the true population.

3. Feature engineering and dimensionality reduction

I add three features. A native region feature groups native country into US, non-US, and Unknown, which turns a column that is about 90 percent United States into the split my fairness audit needs while keeping missing-country people separate. Two binary flags mark whether a person has any capital gain or any capital loss, because those columns are almost always zero with a few very large values, so a flag captures the wealth signal without the model over-reacting to the raw dollar amount. An hours group feature bins hours per week into ranges, which helps the education versus hours comparison in Hypothesis 1 since the effect of hours is not a straight line.

I one-hot encode the categorical columns, which gives a feature matrix of 52 columns, and I scale the numeric features so no single one dominates because of its units. In the real pipeline the scaler is fit on the training split only to avoid leaking information from the test set. A quick PCA on the numeric features spreads the variance fairly evenly across the components, so there is no small set that captures almost everything. Since there are only a handful of numeric features and Hypothesis 1 depends on knowing which original feature matters, reducing them into PCA components would hurt interpretability without buying much, so I keep the original features.

4. Showing the impact of preprocessing

Before cleaning, workclass was 5.64 percent, occupation 5.66 percent, and native country 1.79 percent missing, with 24 exact duplicate rows. After cleaning there are no missing values and none of those duplicates left.

For the class balance, I split the data once with stratification into 26,029 training rows and 6,508 test rows, then apply SMOTE to the training split only. Before SMOTE the training split is imbalanced at about 3.2 to 1, and after SMOTE the minority class is oversampled up to a 1 to 1 balance, while the test set is left untouched so it still reflects the true population.

To show this mattered I fit a default logistic regression before preprocessing and after. Before preprocessing, on the raw imbalanced data, it reaches about 0.815 accuracy but only 0.39 recall on the over 50K class. After scaling, encoding, and oversampling, the accuracy stays about the same at 0.819 but the recall on the over 50K class jumps to 0.79, so the model now catches far more of the real high earners. This is only a sanity check with default settings, but for my lending use case that is the trade I want.

5. Summary statistics

The numeric summary shows age averaging about 38.6, education around 10 years, and hours per week about 40, while capital gain and capital loss have a median of zero and a small mean pulled up by a few large values. The frequency tables show Private is by far the most common workclass, married-civilian-spouse and never-married are the two biggest marital groups, there are roughly twice as many men as women, and the US region dominates.

For outliers I use the IQR rule. It works by sorting the values and finding Q1 and Q3, the 25th and 75th percentiles, so the middle half of the data sits between them. The interquartile range is the width of that middle half, and the rule draws a fence one and a half IQRs beyond each end:

$$\text{IQR} = Q3 - Q1, \text{ fence} = [Q1 - 1.5 \times \text{IQR}, Q3 + 1.5 \times \text{IQR}]$$

Anything outside the fence is flagged. I use this rather than a z-score, $z = (x - \text{mean}) / \text{sd}$, because the mean and standard deviation are themselves dragged around by the extreme values I am trying to find, so on skewed data the z-score fence stretches to accommodate the outliers. Q1 and Q3 do not move no matter how extreme the tail gets, which is what makes the rule robust. Capital gain, capital loss, and hours are all heavily skewed, so a z-score would mislabel a lot of ordinary points.

The rule flags hours per week at 27.7 percent, capital gain at 8.3 percent, capital loss at 4.7 percent, education at 3.7 percent, and age at 0.4 percent.

That is a lot of flagged rows, so the obvious question is whether to drop them. I decided not to, and rather than argue it from principle I measured what dropping them would cost.

Dropping every flagged row would remove 39.7 percent of my data and 47.2 percent of all my high earners. Nearly half the class I am trying to predict. For a use case built on finding high earners, that ends the argument on its own.

The flagged rows are also the signal rather than noise. Rows flagged on capital gain are 61.8 percent high earners against 20.7 percent for everyone else, and they are 21.4 percent of every high earner in the dataset. Capital gain is the strongest feature on permutation importance in section 7.1, so the outliers on my best feature are disproportionately the people I most want to catch.

The hours flag rate is an artifact of the rule, not evidence of bad data, and the formula above shows exactly why. The fence width is set entirely by the middle half of the data, so a narrow middle half gives a narrow fence. 46.7 percent of people work exactly 40 hours, which squeezes the interquartile range to just 5, from a Q1 of 40 to a Q3 of 45. Plugging that in gives a fence of $40 - 7.5$ to $45 + 7.5$, or 32 to 52 hours, so someone working 55 hours gets called an outlier. Working 55 hours a week is not an anomaly, it is a job. The 27.7 percent is measuring the spike at 40, not measuring errors.

Dropping would also bias the fairness audit. Foreign-born people are flagged at 45.3 percent against 39.2 percent for US-born, so removing flagged rows would disproportionately remove people from the smaller group I audit in Hypothesis 2, and I would be measuring the nativity gap on a sample I had already skewed by nativity. As a smaller note, dropping would move the class balance from 24.1 percent high earners to 21.1 percent, making the imbalance worse, which is the opposite of what the rest of my preprocessing is doing.

An IQR flag means a value is far from the middle. It does not mean the value is wrong. These are real people with real recorded incomes, tree models split on order rather than magnitude so the skew does not hurt them, and I keep all of them.

6. Visualizations

The histograms confirm age is roughly bell shaped near 38, education clusters at high school, hours pile up at 40, and capital gain and loss are almost entirely zero. The correlation heatmap shows every off-diagonal correlation is small, so there is no strong multicollinearity, which matters for the logistic regression baseline because correlated inputs make its coefficients unstable and Hypothesis 1 leans on

those coefficients.

The boxplot of education by income shows the over 50K group sits clearly higher on the education scale, though the boxes still overlap. A scatter of age against hours worked shows the over 50K points cluster in middle age and at or above full time hours, while the under 50K points are more common among the young and among people working fewer hours, with heavy overlap between the two, which says no single pair of features cleanly splits them.

7. Patterns tied to my hypotheses

For Hypothesis 1, education produces a spread of 0.741 in the over 50K rate from the lowest to the highest level, while hours produces a spread of only 0.361. Education climbs steadily and cleanly, while hours trends up but flattens at the top. This is a hint in favor of education being the stronger driver, not a final answer, because it is a one-feature-at-a-time view and education is correlated with other features. Section 7.1 is the real test.

For Hypothesis 2, I am showing the base-rate gap the models will later be audited against. US-born people earn over 50K at about 24.6 percent and foreign-born people at about 18.7 percent. If a model just learns that gap, it can end up missing real high earners among foreign-born people, which shows up as lower recall for that group, and that recall gap is what my fairness audit measures for each model. Breaking the rate out by sex shows men earn over 50K at a higher rate in both nativity groups, so nativity is not an isolated effect, which is why the plan audits sex and race alongside it.

For Hypothesis 3, the EDA shows the kind of patterns that flexible models pick up better than a straight linear model. The over 50K rate by age is a clear curve that rises through the working years and falls for the oldest ages, having any capital gain flips the rate from low to very high, which is a sharp threshold effect, and the education by marital status view shows the rate depends on the two features together rather than each one added up. Logistic regression only gets interactions if I add them by hand, while XGBoost and the MLP find them on their own, so the data is consistent with expecting XGBoost to classify better.

7.1 Feature importance from a decision tree

The rate comparison above looks at one feature at a time, so it cannot tell me whether education still matters once every other feature is in the model at once. A decision tree can, because it picks its splits with all the features competing against each other. My professor suggested using a tree for this.

I report two importance numbers instead of one. The tree's built in importance is based on impurity reduction, and that measure is known to favor continuous features. My education column is a single numeric column while my categorical features are split across many one-hot columns, so the built in number could flatter education for the wrong reason. Permutation importance on the held out test rows, scored on PR-AUC, is the cross check.

Measure	Education	Hours worked	Ratio
Tree importance (impurity reduction)	0.223	0.028	8.0x
Permutation importance (PR-AUC)	0.099	0.023	4.3x

Education wins on both, which is what I wanted, because it means the result is not just the impurity measure favoring a numeric column. That is a real point in favor of Hypothesis 1 and a much stronger one than the rate comparison, since every other feature was competing for the same splits. The drop on education from 0.223 to 0.099 between the two measures is that impurity bias appearing where I expected it, which is why I report both rather than picking whichever number I like better.

Two features do rank above education, marital status on the tree at 0.426 and capital gain on permutation importance at 0.201. That does not affect Hypothesis 1, which claims education beats hours worked and not that education leads every feature, but it is worth stating plainly rather than reading the chart as if education came first.

I also fit a shallow tree at depth three, which is too simple to be my real model but is readable on its own. It splits first on marital status, then on education and capital gain, which matches the importance ranking. That is the demonstration behind the explainability note in section 8.

7.2 Statistical tests on the EDA

Everything above is descriptive, so my professor's suggestion to add a t-test is a good one: it puts a test behind the patterns rather than leaving them as eyeballed differences. I picked the tests to match what each feature actually is, since running the wrong test on the wrong shape of data is worse than not testing.

One thing has to be said before any number below, because it changes how all of them read. My sample is 32,537 rows. At that size almost any difference is statistically significant, so a small p-value is close to meaningless on its own. It tells me a difference is probably not zero, which I already assumed. What I want to know is whether the difference is big enough to matter, so I report an effect size next to every test and read that instead.

Welch's t-test on the numeric features. The question: do high earners and everyone else really have different average education, hours, and age, or could a gap that size come from chance?

A t-test does not just check whether two means differ. It scales that difference by how noisy the groups are, because a gap between two wildly spread out groups is easy to get by accident while the same gap between two tight groups is not. The t statistic is that ratio, signal over noise:

$$t = (M1 - M2) / \text{sqrt}(s1^{**}2/n1 + s2^{**}2/n2)$$

where **M1**, **M2** are the two group means, **s1**, **s2** their variances (how spread out each group is), and **n1**, **n2** the group sizes. The top is the difference I care about; the bottom is the standard error of that difference.

Notice what n does on the bottom. As the sample grows, $s^{**}2/n$ shrinks, the denominator shrinks, and t grows. That is exactly why all my p-values are near zero. It is arithmetic, not a discovery.

I use Welch's version rather than the standard one because the standard test assumes both groups have the same variance and pools them into a single estimate. My groups are 7,839 against 24,698 and there is no reason their spreads should match. Welch's keeps **s1** and **s2** separate, as written above, and adjusts the degrees of freedom to compensate.

Cohen's d, which is the number I actually read. Because t grows with sample size, it cannot tell me whether a difference is big. Cohen's d takes the sample size out and asks how many standard deviations apart the two groups are:

$$d = (M1 - M2) / sp, \text{ where } sp = \sqrt{((n1-1)*s1^{**2} + (n2-1)*s2^{**2}) / (n1+n2-2)}$$

where **sp** is the pooled standard deviation, a weighted average of both groups' spread. There is no *n* dividing the difference, so *d* does not inflate with more rows. The usual convention is 0.2 small, 0.5 medium, 0.8 large.

Feature	Mean >50K	Mean <=50K	Cohen's d	Interpretation
education_num	11.61	9.60	0.832	Large effect
age	44.25	36.79	0.563	Medium effect
hours_per_week	45.47	38.84	0.552	Medium effect

Every p-value is effectively zero, which is the sample size talking and is not the interesting part. The effect sizes are. Education is a large effect and hours is a medium one, so the gap between high earners and everyone else is substantially wider on education. That is the third independent way I have gotten the same answer, after the rate spread in section 7 and the decision tree in section 7.1. Three different methods, three times education ahead of hours, is much better support for Hypothesis 1 than any one of them alone.

I checked the t-test's assumption rather than taking it on faith. A t-test needs the group means to be roughly normally distributed, and with samples this large the central limit theorem delivers that for mildly skewed features regardless of the underlying shape. Education, hours, and age are skewed only -0.31, 0.23, and 0.56, so the test is safe there. Capital gain has skew 11.95 and excess kurtosis 154.66, which is far enough from normal that I do not trust a t-test on it, and it is the same fact that made me use the IQR rule instead of z-scores in section 5.

Mann-Whitney U where the t-test does not apply. Instead of using the values, this test throws them away and keeps only their order. Pool both groups, rank every value from smallest to largest, and add up the ranks belonging to group 1:

$$U1 = n1*n2 + n1*(n1+1)/2 - R1$$

where **R1** is the sum of the ranks in group 1. In plain terms, U counts how often a randomly picked value from one group beats a randomly picked value from the other. If the groups were interchangeable that count would land near $n1*n2/2$, and how far it sits from there is the evidence.

Because it uses only ranks, a capital gain of 99,999 is simply "the biggest one" rather than a number that can drag a mean around. That is why it works where the t-test fails. Its effect size is the rank-biserial correlation:

$$r = 1 - 2*U / (n1*n2)$$

which runs from 0 to 1 and reads as how lopsided the win rate is between the two groups.

Education comes back at 0.434 against 0.344 for hours, the same ordering the t-test found. The two agreeing means the Hypothesis 1 result does not depend on the t-test's normality assumption, which is the whole point of running both.

Chi-square for the categorical features. A t-test compares means, and there is no mean of "occupation," so it cannot touch most of my feature set. Chi-square asks a different question: does the income split depend on the category at all?

It works by building a table of counts for each category-by-income cell, then working out what that table would look like if the two were completely unrelated. If 24.1 percent of everyone earns over 50K, then under independence 24.1 percent of every occupation should too. That gives an expected count per cell:

$$E = (\text{row total} \times \text{column total}) / N$$

Chi-square then measures how far the real table sits from that imaginary independent one, summed over every cell:

$$\text{chi-squared} = \text{SUM } (O - E)^2 / E$$

where **O** is the observed count and **E** the expected count. Squaring makes over- and under-shoots both count. Dividing by **E** is the important part: being 50 people off matters enormously in a cell expecting 60, and barely at all in one expecting 10,000.

Chi-square inflates with sample size the same way *t* does, so it needs an effect size too. Cramer's *V* rescales it to a 0 to 1 number:

$$V = \sqrt{\text{chi-squared} / (N \times (k - 1))}, \quad k = \min(\text{rows}, \text{columns})$$

Dividing by **N** removes the sample-size inflation. Dividing by **k-1** stops features with more categories from scoring higher just for having more cells to accumulate error in, which matters to me because occupation has 15 categories and sex has 2 and I want to compare them fairly.

Feature	Cramer's V	p
relationship	0.454	<0.001
marital_status	0.447	<0.001
occupation	0.352	<0.001
sex	0.216	<0.001
workclass	0.179	<0.001
race	0.101	<0.001
native_region	0.039	2.5e-11

The marital and relationship features have the strongest association with income, which lines up with the decision tree in section 7.1 where marital status was also the top feature. Two different methods seeing the same structure is a good sign.

Native region is the row worth pointing at. Its p-value is 2.5e-11, which looks impressive, and its Cramer's *V* is 0.039, which is essentially no association. That one row is the clearest example of why I am not reading p-values here. Judged by significance alone I would call nativity an important predictor of income. Judged by effect size it is the weakest categorical feature I have. Both are true, and the effect size is the one that means something.

A two-proportion z-test on the Hypothesis 2 base-rate gap. Hypothesis 2 assumes there is a real base-rate difference across nativity for a model to learn. Before building a hypothesis on that I should check it is not sampling noise, especially since the foreign-born group is only 2,802 people.

This test has the same shape as the *t*-test, a difference divided by its own uncertainty, but for proportions rather than means. For a proportion the spread is not measured separately, it follows from the proportion

itself, since a rate p has variance $p(1-p)$. So the standard error comes from the rates and group sizes alone:

$$z = (p_1 - p_2) / \sqrt{p_{\text{pooled}} * (1 - p_{\text{pooled}}) * (1/n_1 + 1/n_2)}, \quad p_{\text{pooled}} = (x_1 + x_2) / (n_1 + n_2)$$

where p_1, p_2 are the two observed rates, x_1, x_2 the counts of high earners in each group, and p_{pooled} the combined rate. The pooled rate is used because the null hypothesis says both groups share one true rate, so under the null that is the best estimate of it.

US-born earn over 50K at 24.59 percent ($n = 29,153$) against 18.70 percent for foreign-born ($n = 2,802$). The gap is 5.9 percentage points, $z = 6.97$, $p = 3.3\text{e-}12$.

The confidence interval is what I actually report, because the z and its p -value only say the gap is probably not zero, while the interval says how big it plausibly is:

$$(p_1 - p_2) \pm 1.96 * \sqrt{p_1(1-p_1)/n_1 + p_2(1-p_2)/n_2}$$

This one does not pool, because here I am estimating the size of a real difference rather than testing whether it is zero, so each group's own rate is the right thing to use. The **1.96** is the z value that puts 95 percent of a normal distribution inside the interval.

That gives 4.4 to 7.4 percentage points. The interval excludes zero, so the gap is real and not sampling noise, and it also tells me how big it plausibly is.

This is worth holding next to the Cramer's V of 0.039 above. They are both about nativity and they do not conflict: the gap is real and the gap is small. That is the honest reading, and it matters for how I interpret Hypothesis 2 later, because if a model produces a recall gap I cannot immediately blame the model when some of it is this base rate coming through. What Hypothesis 2 actually asks is which of the three models handles that pre-existing gap best.

What these tests do not do. They confirm the EDA rather than settling anything. None of my three hypotheses are answered here. Hypothesis 1 is about a trained model's feature importance, not group means. Hypotheses 2 and 3 are comparisons between models, which need paired tests on the same held-out rows, and my plan uses McNemar's test and bootstrap confidence intervals for those. A t -test would be the wrong tool there, because the three models predict on the same people, so their errors are paired rather than independent, and treating them as two independent samples would overstate my confidence. These tests belong to the EDA, which is where I have kept them.

8. Explainability, for the final report's model comparison

My professor asked for a brief note on explainability in the model comparison section of the final report. The decision tree work in section 7.1 is the groundwork for it, and the note I plan to write is a three way contrast.

Logistic regression is transparent. Each feature has one signed coefficient, so the direction and size of its effect are readable straight off the model. The limit is that it only sees additive effects, so anything it cannot express it also cannot explain.

XGBoost is opaque on its own, since no single tree carries the answer, but it is explainable from the outside. Gain importance ranks the features, and a shallow tree fit on XGBoost's predictions rather than on the true labels gives a readable approximation of its decision path. That surrogate tree is what I will show,

and section 7.1 already fits the same kind of tree against the true labels as a demonstration: it splits first on marital status, then on education and capital gain, which matches the importance ranking.

The MLP has no native explanation at all. Its weights are not interpretable in any useful way, so permutation importance applied from the outside is the only option, and it tells me which features matter without telling me how.

The pattern is that explainability drops as flexibility rises, which is the trade Hypothesis 3 is really about. If XGBoost wins on PR-AUC and F1 by a small margin, the interpretability cost is worth weighing against it, especially for the lending use case that frames this project, where a decision a person cannot have explained to them is a problem regardless of how accurate it is.

9. Summary

Hypothesis 1 gets a real point in its favor. Education produces a wider and cleaner spread in the over 50K rate than hours worked, and once every feature competes in the same decision tree, education still beats hours on both importance measures, about eight times on tree importance and four times on permutation importance. That is to be confirmed against the coefficients and permutation importance on the trained models.

Hypothesis 2 shows a real base-rate gap across nativity that holds up alongside sex, so the models will have something to learn and the audit will have something to measure.

Hypothesis 3 sees non-linear, interaction and threshold effects, the kind of structure flexible models can use, which is consistent with expecting XGBoost to classify better.

I also looked for anything that cut against these three and nothing did. The main caution is the heavy class overlap on any single feature, which limits how strong each hint is but does not point any hypothesis the other way.

How I used an LLM for this assignment

I used an LLM to help create some of the visualizations in this notebook and also to help create some markdown cells in this notebook so I could correctly interpret some of the visualizations in this notebook too. I wrote and edited the analysis myself and confirmed the numbers by running the notebook top to bottom.

For this revision I used an LLM to work through my professor's feedback about using a decision tree for feature importance, including why the tree's built in importance can be misleading on one hot encoded features and why permutation importance is worth reporting alongside it. I also used it to think through which statistical tests actually fit my data, in particular why a t-test is fine on education and hours but not on capital gain, and why effect sizes rather than p-values are the thing to read at my sample size.

I finally used an LLM to confirm that I covered every part of the instructions for this notebook to ensure it is fully complete.